Questions by topic: Aquaculture - Set 1 (65 marks)

Answer all questions in the spaces provided.

0 1 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

0 1 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

0 1 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

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Dissolved oxygen .....

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Water flow .....

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Day length .....

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Temperature .....

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0 2 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 2 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

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0 2 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 2 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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0 3 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed .....

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Sea lice and disease .....

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Escapes affecting wild gene pools .....

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Organic waste pollution .....

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0 3 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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0 4 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 4 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 4 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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0 5 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

0 5 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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0 6 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 6 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

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Answer

0 6 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

0 6 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

0 6 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

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0 6 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

15

END OF QUESTIONS